THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 10: **Tissues in Action** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Xylem vessels are dead cells yet efficient in conducting water. Reason (R): Their hollow lignified walls form continuous tubes. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The secondary xylem is produced by: 1

(a) apical meristem **(b)** vascular cambium towards the inside
(c) intercalary meristem **(d)** cork cambium

3. Skeletal muscle fibres are multinucleate because: 1

(a) they are neurons **(b)** they have no cells
(c) fusion of many cells during development **(d)** they are small

4. The gap between two neurons is called the: 1

(a) node **(b)** axon
(c) synapse **(d)** cleft in the skull

5. Iodine stain is used to detect: 1

(a) starch in parenchyma **(b)** protein
(c) lignin **(d)** cellulose

6. Assertion (A): Phloem is a living tissue. Reason (R): Sieve tubes need companion cells for their functioning. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. The cork cambium produces cork towards the: 1

(a) centre **(b)** outside **(c)** inside **(d)** pith

8. Which tissue would you expect to find in the highest proportion in the wall of the heart? 1

(a) cartilage **(b)** bone **(c)** adipose **(d)** cardiac muscle

9. Neurons cannot divide in adults. This means: 1

(a) all neurons are dead
(b) nerves regenerate quickly
(c) nerves are cells without nucleus
(d) damaged nerve cells are not easily replaced

10. A student cuts a stem and places it in red ink for some time. The red colour will be seen in the: 1

(a) xylem **(b)** phloem **(c)** cork **(d)** epidermis

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the importance of the cuticle and the stomata in a desert plant? 2

12. Explain why red blood cells and white blood cells are produced continuously. 2

13. Why are sieve tubes dependent on companion cells although they are living? 2

14. Why is a fracture of a bone slower to heal than a skin cut? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compare the structures of the three types of muscle with the help of their arrangement of nuclei and striations. 3

16. A section of a tissue shows cells with a large central vacuole, a thin wall and chloroplasts. Identify the tissue and its function. 3

17. Describe how plant fibres such as jute, flax and cotton are obtained and which tissue they come from. 3

OR

Why does a tree bleed a sticky liquid when its bark is cut but not a clear water-like liquid from the wood? 3

18. Explain how the organisation of tissues in the skin helps protection, sensation and temperature control. 3

OR

Why do older leaves and bark have more sclerenchyma and why is this tissue important in timber? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how tissues form organs and organ systems with the example of the human heart, listing the tissues and their roles. 5

OR

Explain how a section of a muscle can be prepared and stained to see the striations and nuclei, and describe the differences you expect in the three types of muscle under the microscope. 5

20. Explain the differences between the types of neuron: sensory, relay and motor with the pathway of a reflex, giving the example of a pin prick. 5

OR

Discuss the contribution of tissue culture, with the work of F. C. Steward, in conserving rare plants and producing food, with the steps and the media used. 5

END OF QUESTION PAPER — Total: 40 Marks